Audio Visualiser DSP Board

https://github.com/drewsum/audio_visualiser

01. Table of Contents

02. 12V Power Input — 12V Power Input — File: POS12_power_input.kicad_sch

03. 12V Telemetry — 12V Telemetry — File: POS12_telemetry.kicad_sch

04. 3.3V Power Supply — 3.3V Power Supply — File: POS3P3_power_supply.kicad_sch

05. 3.3V Telemetry — 3.3V Telemetry — File: POS3P3_telemetry.kicad_sch

06. dsPIC33 Programming — dsPIC33 Programming — File: dspic33_programming.kicad_sch

07. dsPIC33 Power — dsPIC33 Power — File: dspic33_power.kicad_sch

08. dsPIC33 GPIO — dsPIC33 GPIO — File: dspic33_gpio.kicad_sch

09. USB UART Bridge — USB UART Bridge — File: usb_uart_bridge.kicad_sch

10. Platform ETC — Platform ETC — File: platform_etc.kicad_sch

11. PGOOD LEDs — PGOOD LEDs — File: pgood_leds.kicad_sch

12. Status LEDs — Status LEDs — File: status_leds.kicad_sch

13. RCA Signal Conditioning — RCA Signal Conditioning — File: rca_signal_conditioning.kicad_sch

14. Speaker Signal Conditioning — Speaker Signal Conditioning — File: speaker_signal_conditoning.kicad_sch

15. Source Selection, PGAs — Source Selection, PGAs — File: source_selection_pgas.kicad_sch

16. Anti-alias Filtering — Anti-alias Filtering — File: anti_alias_filtering.kicad_sch

17. Audio Voltage References — Audio Voltage References — File: audio_voltage_references.kicad_sch

18. LED Panel Buffers — LED Panel Buffers — File: led_panel_buffers.kicad_sch

19. Pushbuttons — Pushbuttons — File: pushbuttons.kicad_sch

20. Mechanical — Mechanical — File: mechanical.kicad_sch

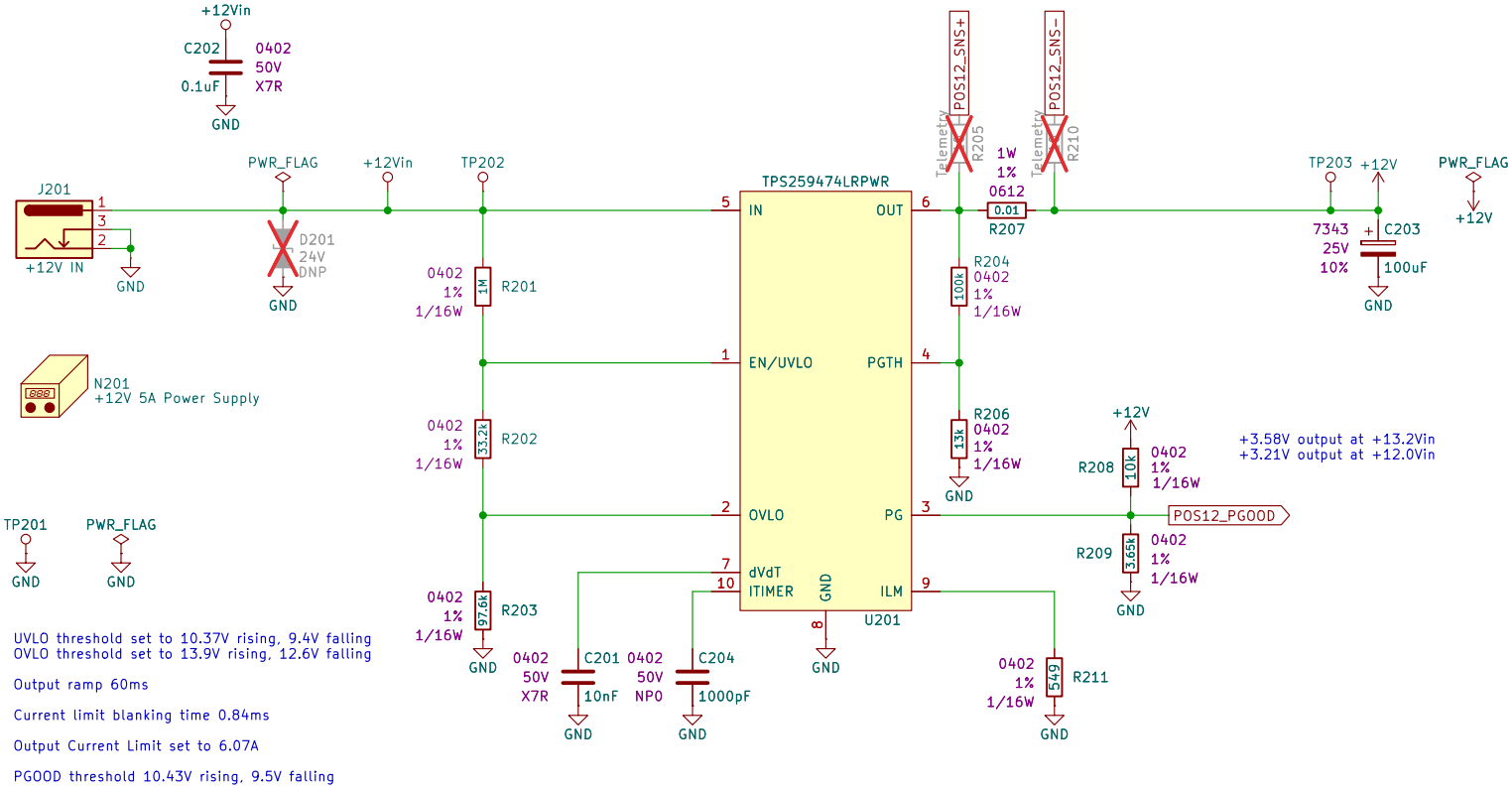
21. LED Panel Connectors — LED Panel Connectors — File: led_panel_connectors.kicad_sch

Drew Maatman and Drew Koetting

Sheet: /
File: Audio_Visualiser_DSP.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.0.7		Id: 1/21



Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A 8 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

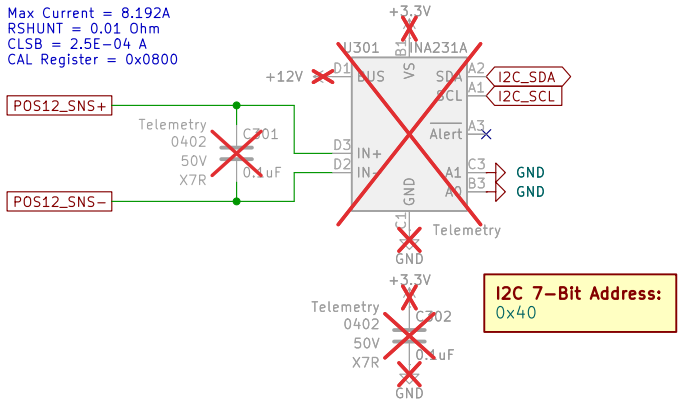
Drew Maatman and Drew Koetting

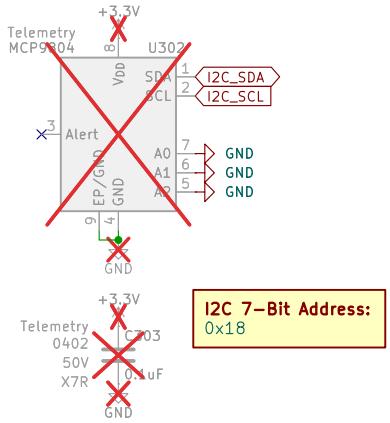
Sheet: /12V Power Input/

File: POS12_power_input.kicad_sch

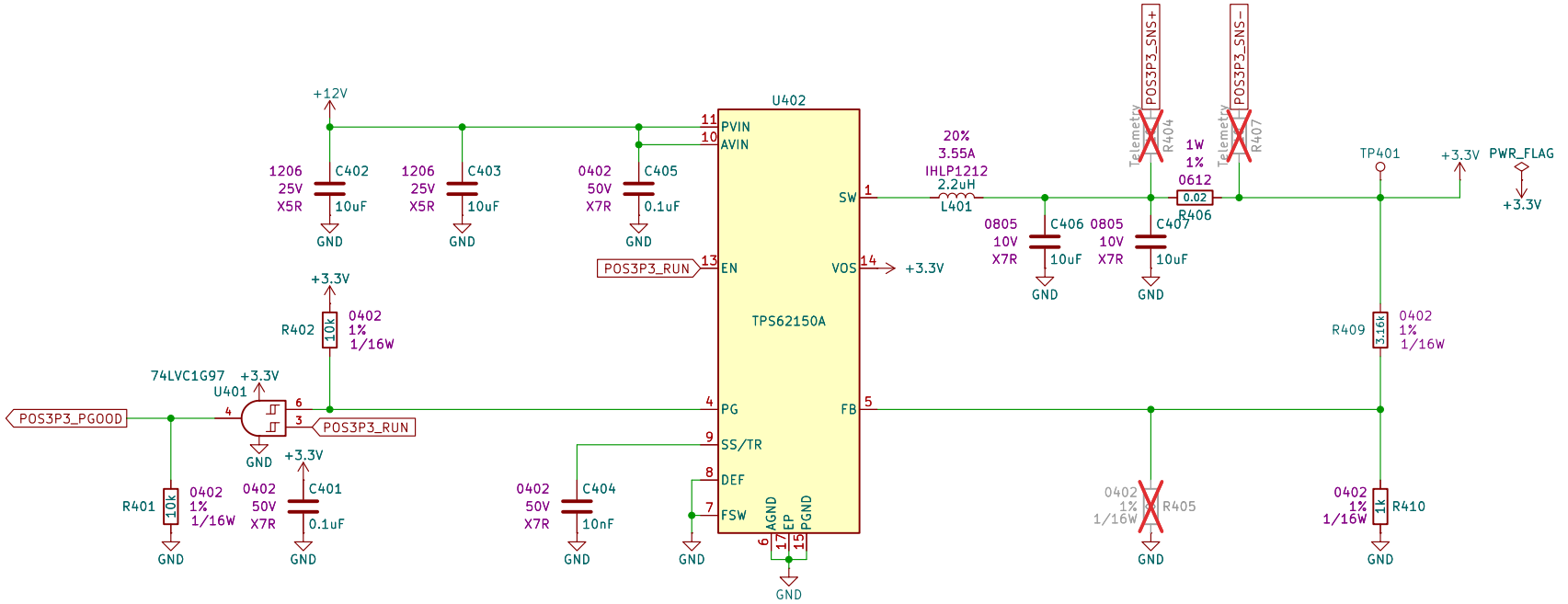
Title: Audio Visualiser DSP

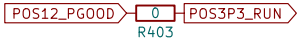
Size: A	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.0.7		Id: 2/21

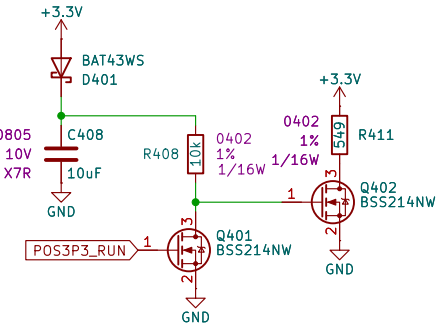




Drew Maatman and Drew Koetting		
Sheet: /12V Telemetry/		
File: POS12_telemetry.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/21







Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	4	4.4mA
MCP6024	5.4mA	3	16.2mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	1	0.2mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffer (TCA9801)	2.9mA	1	2.9mA
I2C Translator (LTC4317)	2mA	1	2mA
Octal Buffers	1mA	2	2mA
LVC1G97s	0.01mA	20	0.2mA
Rotary Encoder Module	50mA	1	50mA
LEDs	1.3mA	9	11.7mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.497W	150.669mA

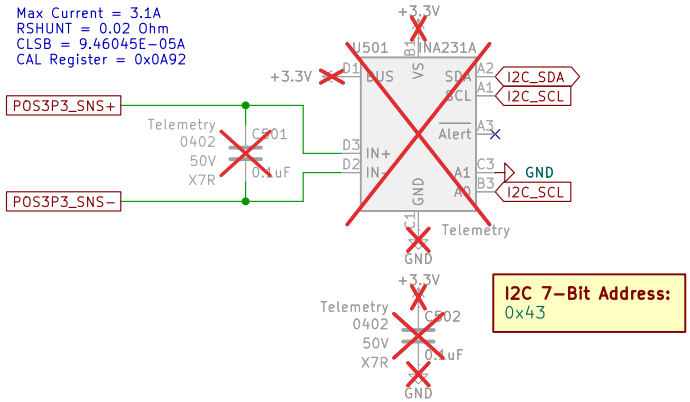
Drew Maatman and Drew Koetting

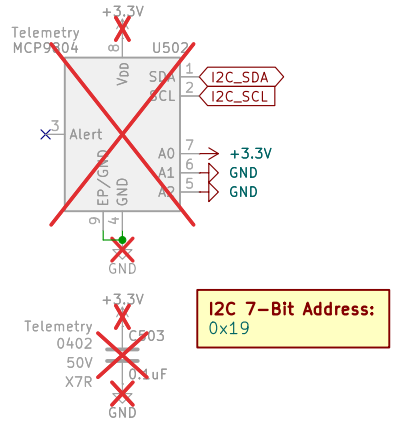
Sheet: /3.3V Power Supply/

File: POS3P3_power_supply.kicad_sch

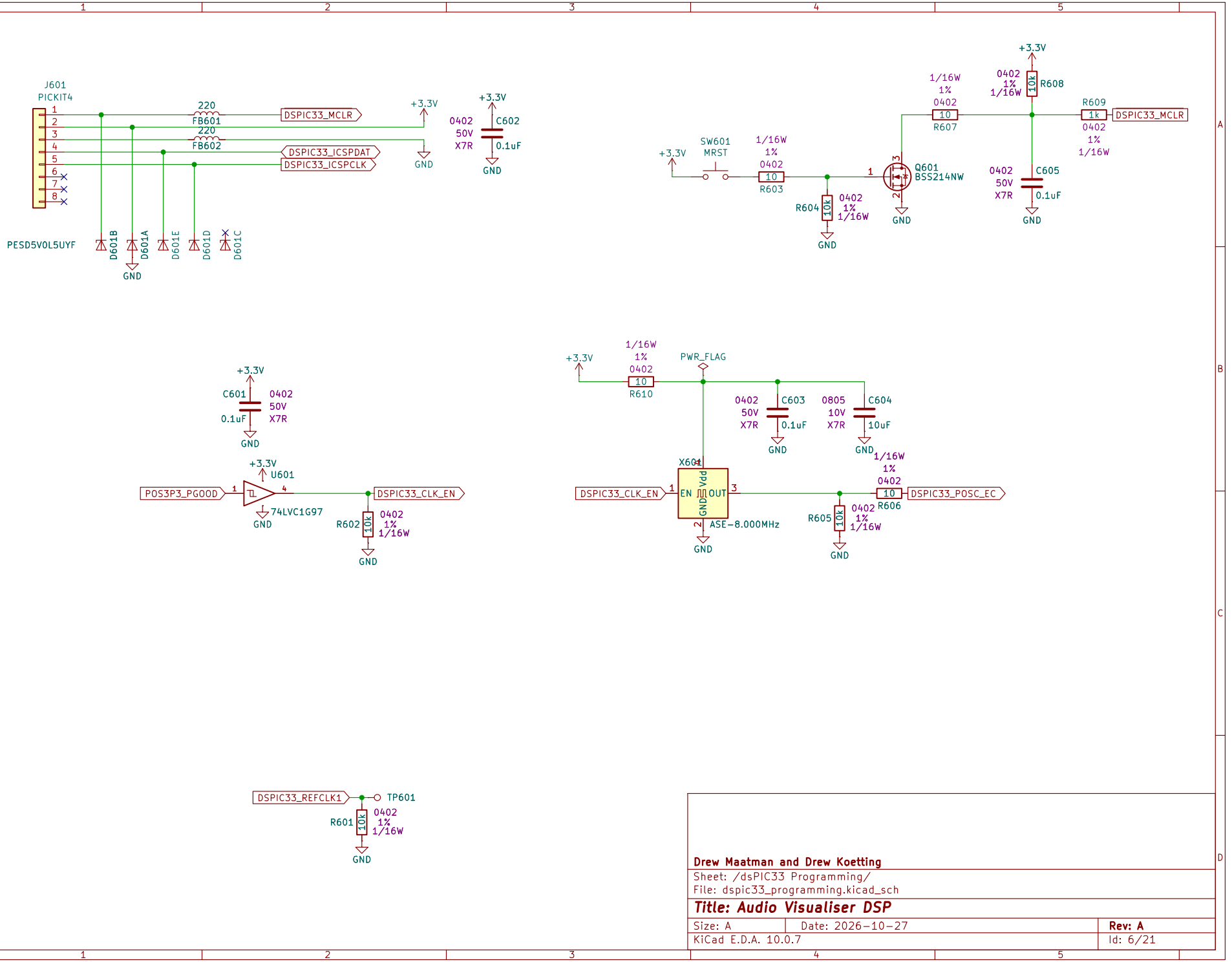
Title: Audio Visualiser DSP

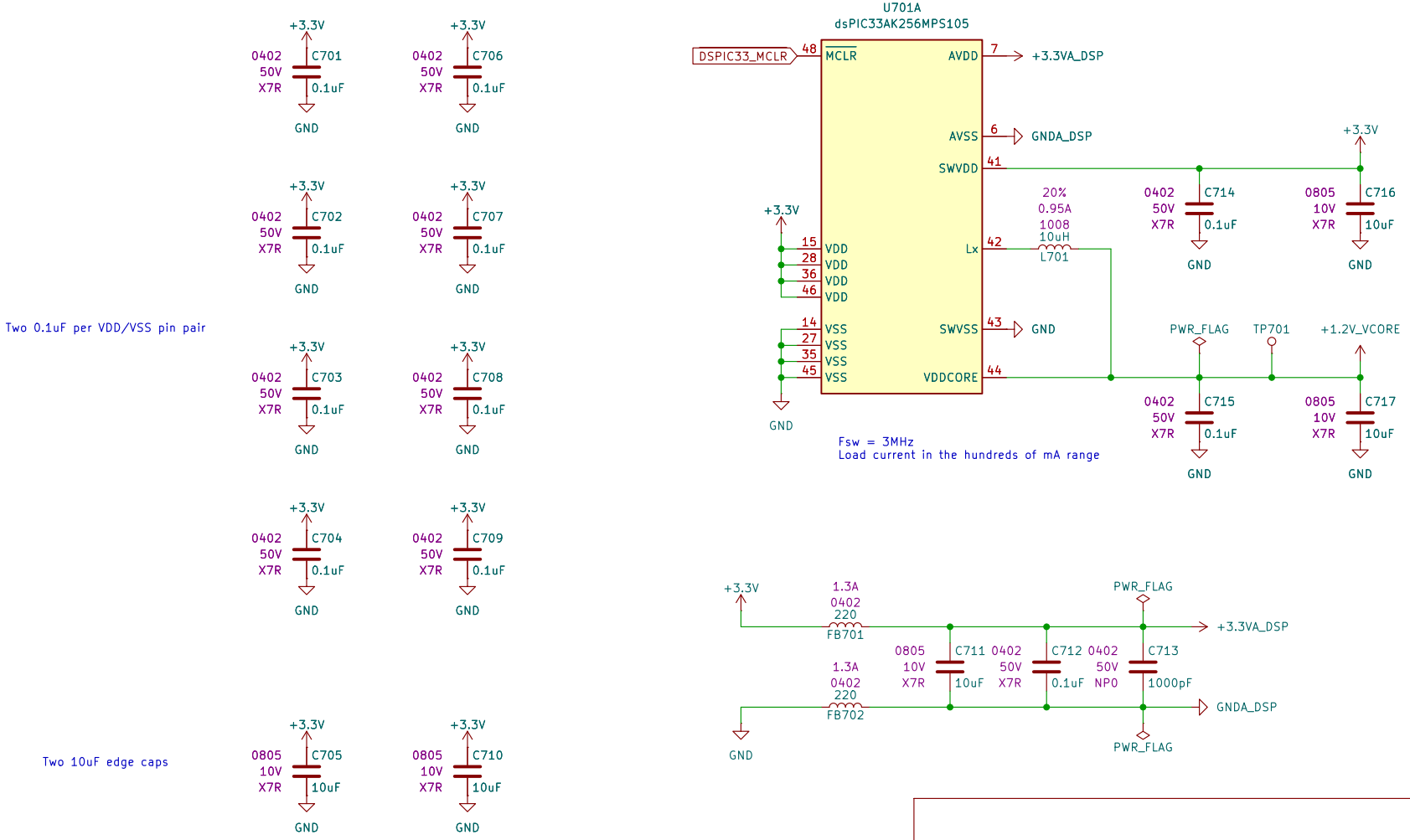
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/21



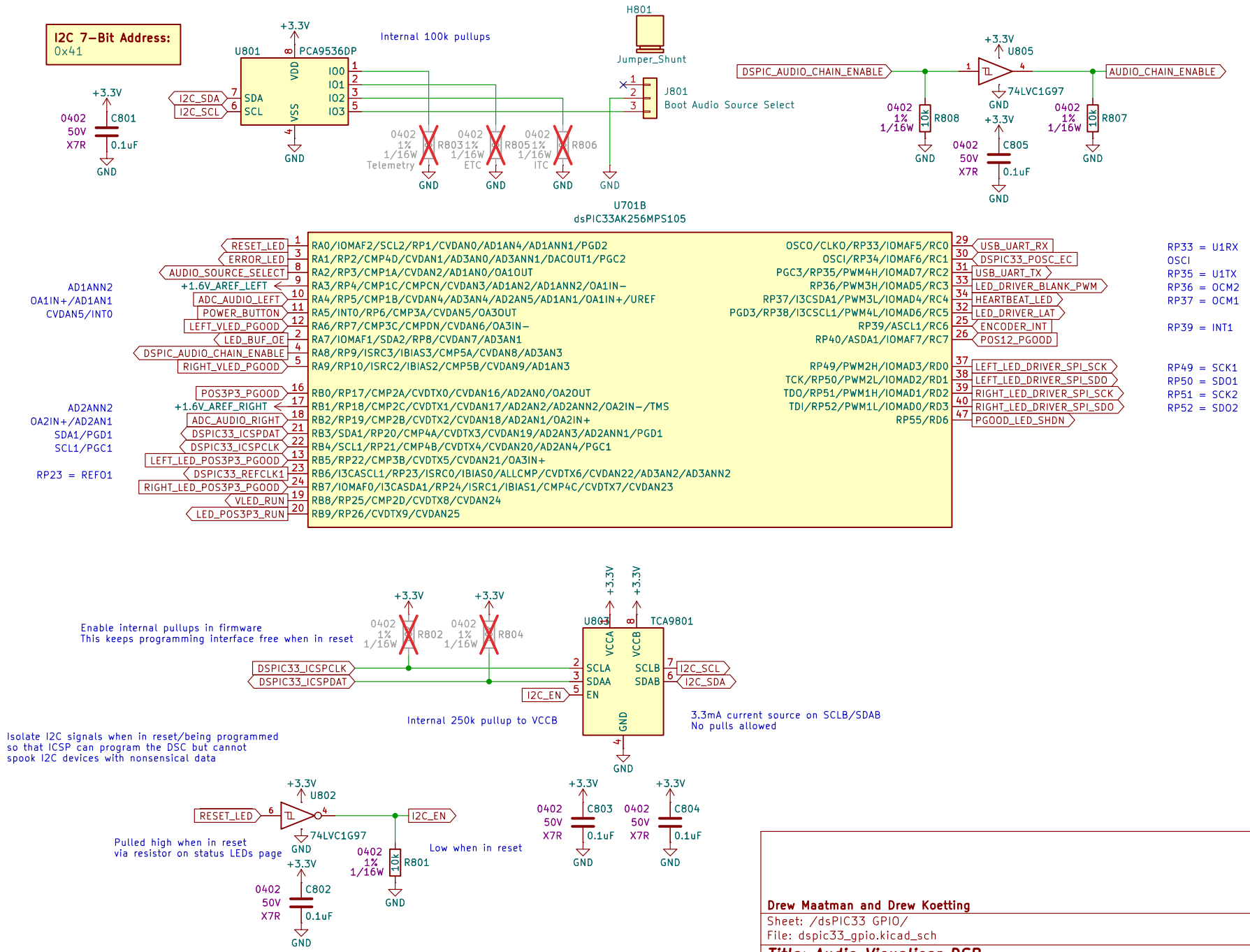


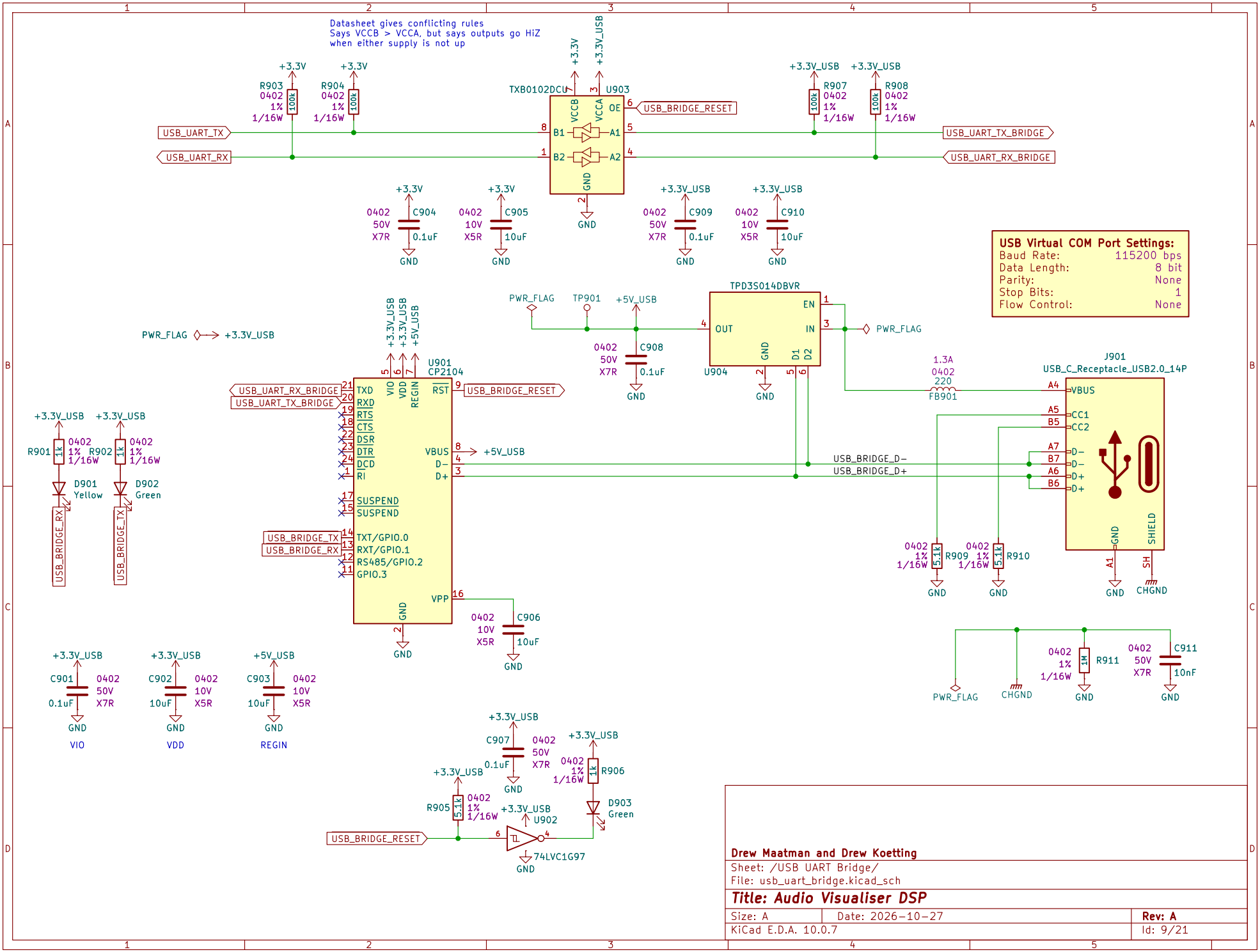
Drew Maatman and Drew Koetting		
Sheet: /3.3V Telemetry/		
File: POS3P3_telemetry.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/21

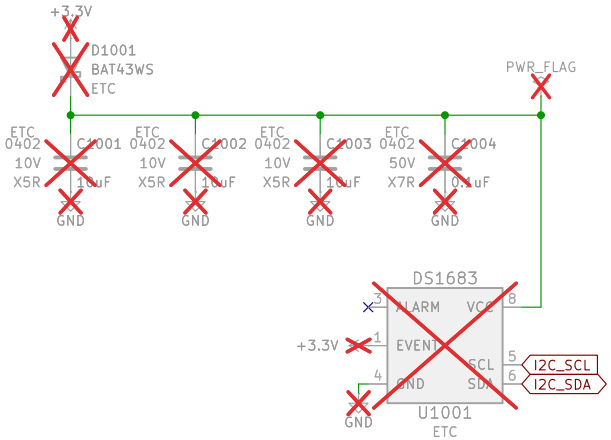

J601
PICKIT4
1
2
3
4
5
6
7
8
220
FB601
220
FB602
DSPIC33_MCLR
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
+3.3V
GND
+3.3V
0402
50V
X7R
C602
0.1uF
GND
PESD5V0L5UYF
D601B
D601A
D601E
D601D
D601C
GND
+3.3V
1/16W
1%
0402
10
R607
0402
1%
1/16W
10k
R608
R609
1k
DSPIC33_MCLR
0402
1%
1/16W
+3.3V
SW601
MRST
1/16W
1%
0402
10
R603
Q601
BSS214NW
R604
10k
0402
1%
1/16W
GND
0402
50V
X7R
C605
0.1uF
GND
+3.3V
C601
0402
50V
X7R
0.1uF
GND
+3.3V
U601
POS3P3_PGOOD
1
TL
4
DSPIC33_CLK_EN
74LVC1G97
GND
R602
10k
0402
1%
1/16W
GND
+3.3V
1/16W
1%
0402
10
R610
PWR_FLAG
0402
50V
X7R
C603
0.1uF
GND
0805
10V
X7R
C604
10uF
GND
X601
DSPIC33_CLK_EN
1
EN
Vdd
OUT
3
GND
2
ASE-8.000MHz
GND
1/16W
1%
0402
10
R606
DSPIC33_POSC_EC
R605
10k
0402
1%
1/16W
GND
DSPIC33_REFCLK1
TP601
R601
10k
0402
1%
1/16W
GND
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 6/21

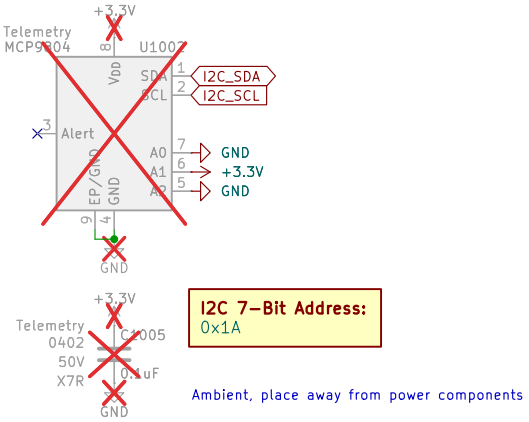


Drew Maatman and Drew Koetting		
Sheet: /dsPIC33 Power/		
File: dspic33_power.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/21


I2C 7-Bit Address:
0x41
U801
PCA9536DP
Internal 100k pullups
H801
Jumper_Shunt
J801
Boot Audio Source Select
R803
Telemetry
R805
ETC
R806
ITC
C801
0.1uF
U805
74LVC1G97
R808
R807
C805
DSPIC_AUDIO_CHAIN_ENABLE
AUDIO_CHAIN_ENABLE
U701B
dsPIC33AK256MPS105
RP33 = U1RX
OSCI
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1
RP39 = INT1
RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2
RP23 = REFO1
Enable internal pullups in firmware
This keeps programming interface free when in reset
U803
TCA9801
R802
R804
Internal 250k pullup to VCCB
3.3mA current source on SCLB/SDAB
No pulls allowed
Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data
U802
Pulled high when in reset
via resistor on status LEDs page
Low when in reset
R801
C802
C803
C804
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 8/21

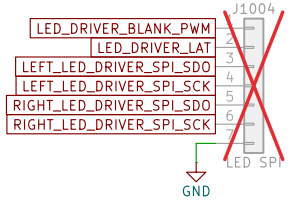

Datasheet gives conflicting rules
Says VCCB > VCCA, but says outputs go HiZ
when either supply is not up
TXB0102DCUR
U903
USB_BRIDGE_RESET
USB_UART_TX
USB_UART_RX
USB_UART_TX_BRIDGE
USB_UART_RX_BRIDGE
R903 0402 1% 1/16W 100k
R904 0402 1% 1/16W 100k
R907 0402 1% 1/16W 100k
R908 0402 1% 1/16W 100k
C904 0402 50V X7R 0.1uF
C905 0402 10V X5R 10uF
C909 0402 50V X7R 0.1uF
C910 0402 10V X5R 10uF
USB Virtual COM Port Settings:
Baud Rate: 115200 bps
Data Length: 8 bit
Parity: None
Stop Bits: 1
Flow Control: None
PWR_FLAG ⟷ +3.3V_USB
TPD3S014DBVR
U904
TP901
C908 0402 50V X7R 0.1uF
FB901 1.3A 0402 220
J901
USB_C_Receptacle_USB2.0_14P
U901
CP2104
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
GPIO.3
R901 1k 0402 1%
R902 1k 0402 1%
D901 Yellow
D902 Green
USB_BRIDGE_RX
USB_BRIDGE_TX
R909 5.1k 0402 1% 1/16W
R910 5.1k 0402 1% 1/16W
C901 0402 50V X7R 0.1uF
C902 0402 10V X5R 10uF
C903 0402 10V X5R 10uF
C906 0402 10V X5R 10uF
VIO
VDD
REGIN
C907 0402 50V X7R 0.1uF
R905 5.1k 0402 1% 1/16W
R906 1k 0402 1% 1/16W
D903 Green
U902
74LVC1G97
R911 1M 0402 1% 1/16W
C911 0402 50V X7R 10nF
CHGND
Drew Maatman and Drew Koetting
Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
KiCad E.D.A. 10.0.7
Id: 9/21





Test Points



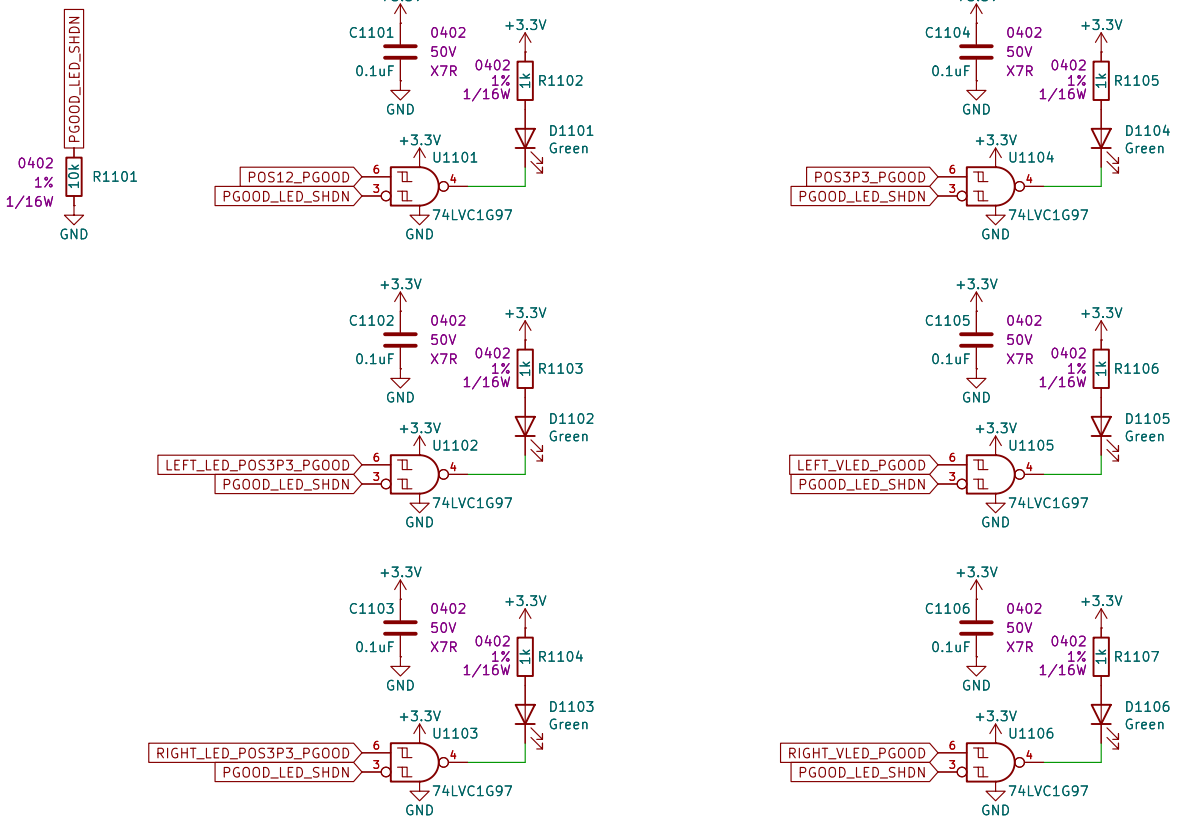


Drew Maatman and Drew Koetting

Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 10/21

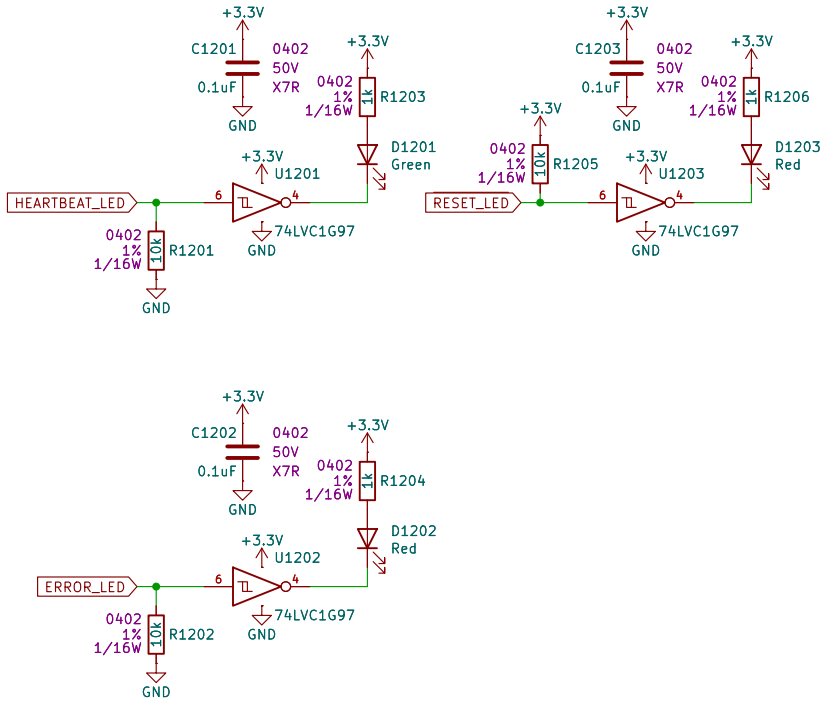


Drew Maatman and Drew Koetting

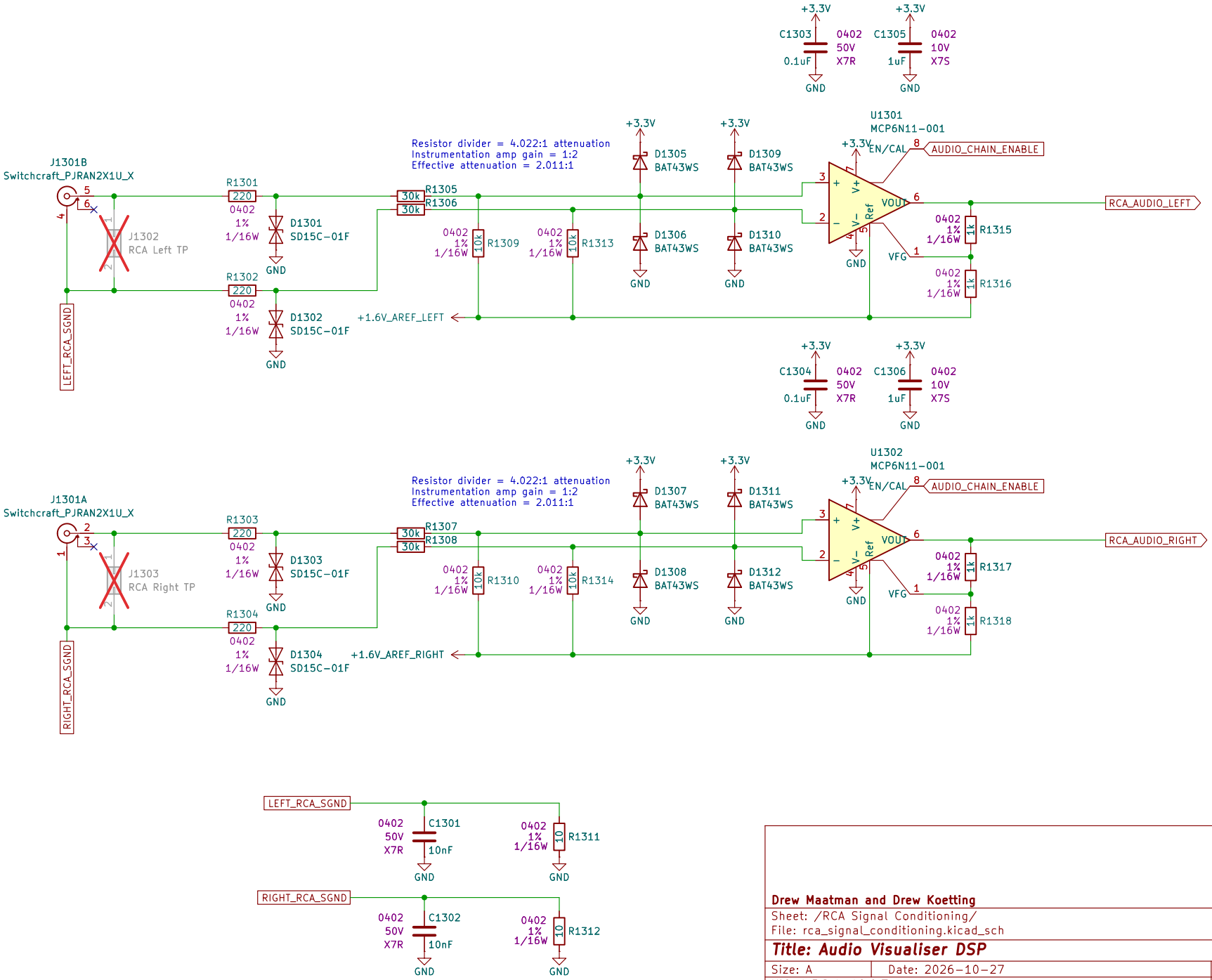
Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 11/21

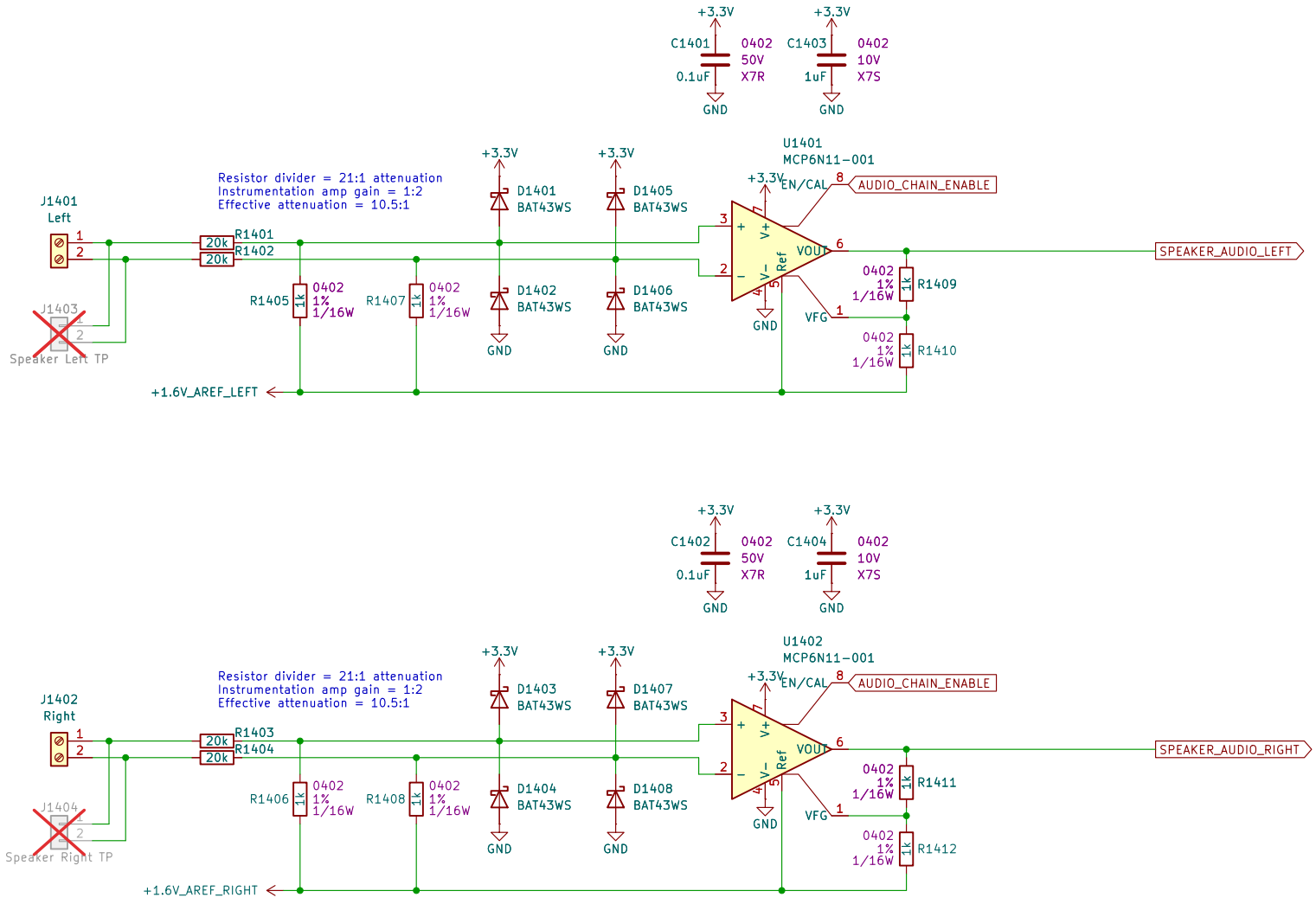


Drew Maatman and Drew Koetting		
Sheet: /Status LEDs/		
File: status_leds.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 12/21

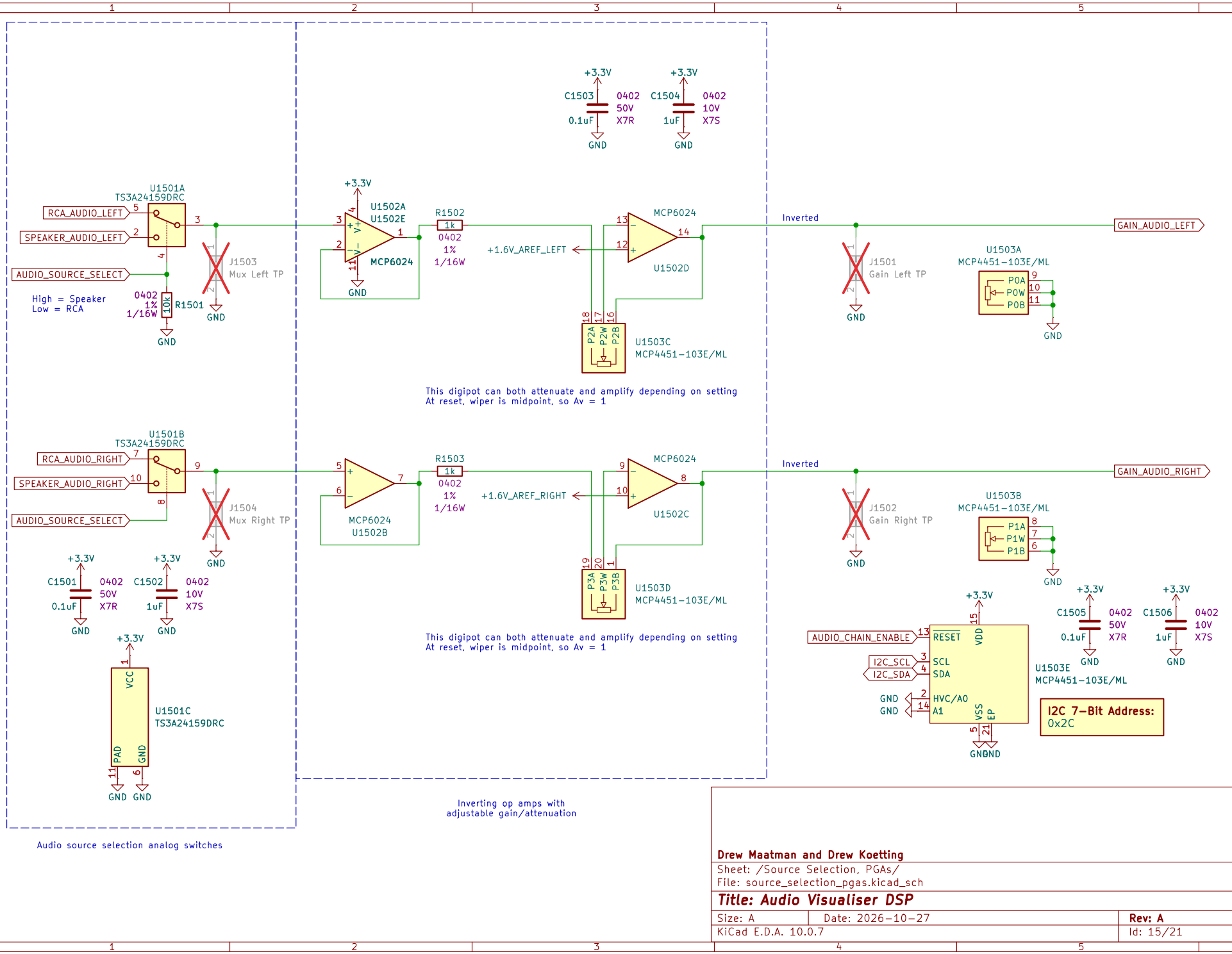


Drew Maatman and Drew Koetting	
Sheet: /RCA Signal Conditioning/	
File: rca_signal_conditioning.kicad_sch	
Title: Audio Visualiser DSP	
Size: A	Date: 2026-10-27
KiCad E.D.A. 10.0.7	Id: 13/21

Rev: A



Drew Maatman and Drew Koetting		
Sheet: /Speaker Signal Conditioning/		
File: speaker_signal_conditoning.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 14/21



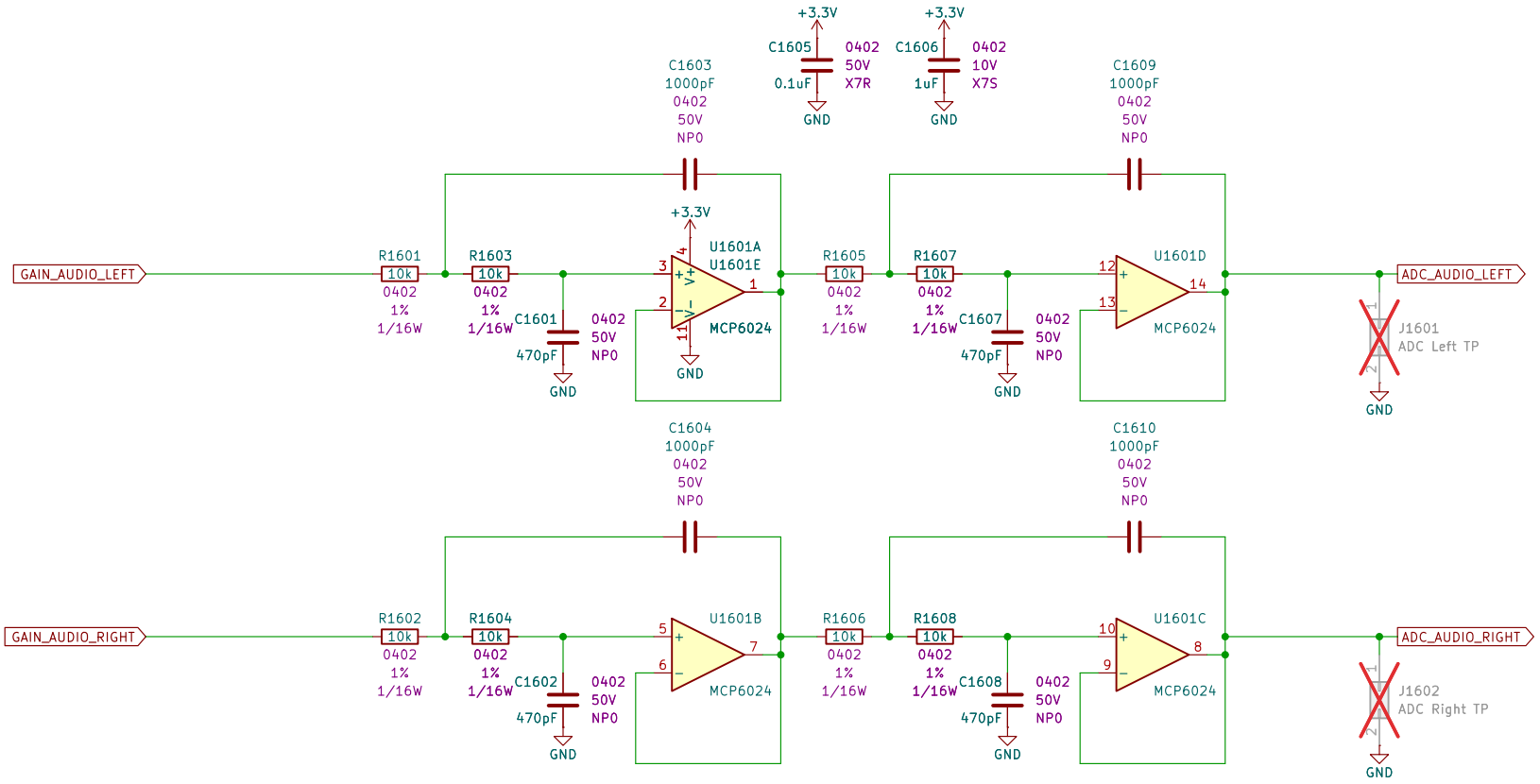
Drew Maatman and Drew Koetting

Sheet: /Source Selection, PGAs/

File: source_selection_pgas.kicad_sch

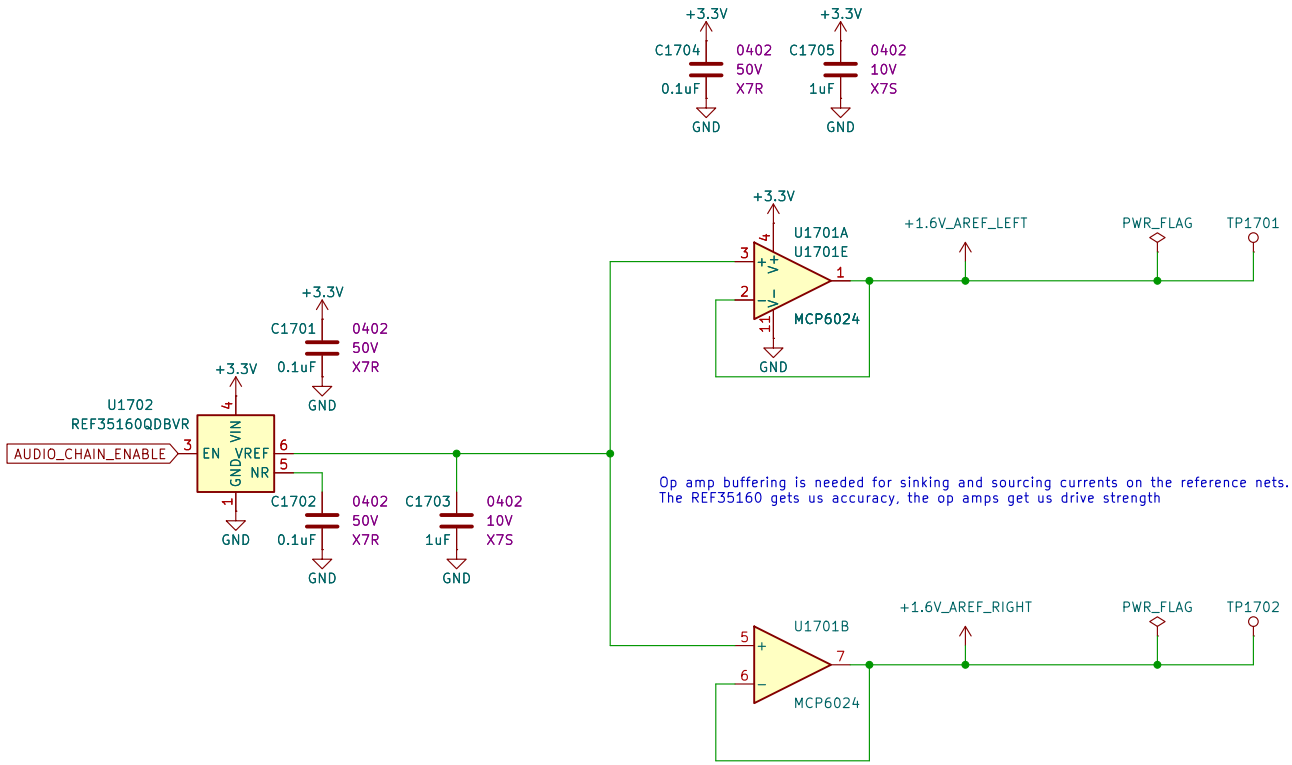
Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.0.7		Id: 15/21



4-pole Sallen-Key active anti-aliasing lowpass filters
Unity gain
fc = 23.2kHz, Q = 0.707 each
If sampling at 44.1kHz, nyquist is 22.05kHz
Must filter hard around this frequency

Drew Maatman and Drew Koetting		
Sheet: /Anti-alias Filtering/		
File: anti_alias_filtering.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 16/21

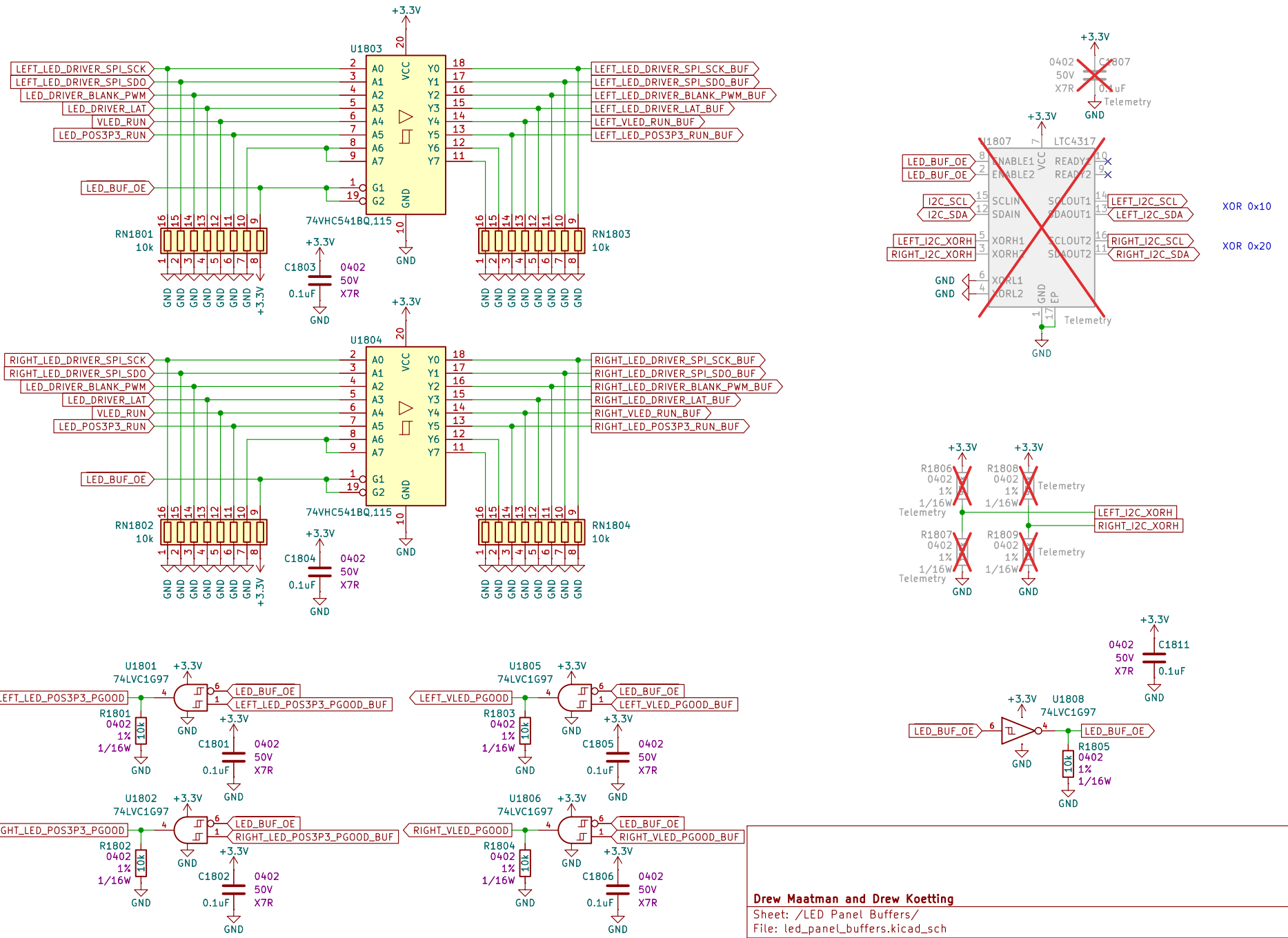


Drew Maatman and Drew Koetting

Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 17/21



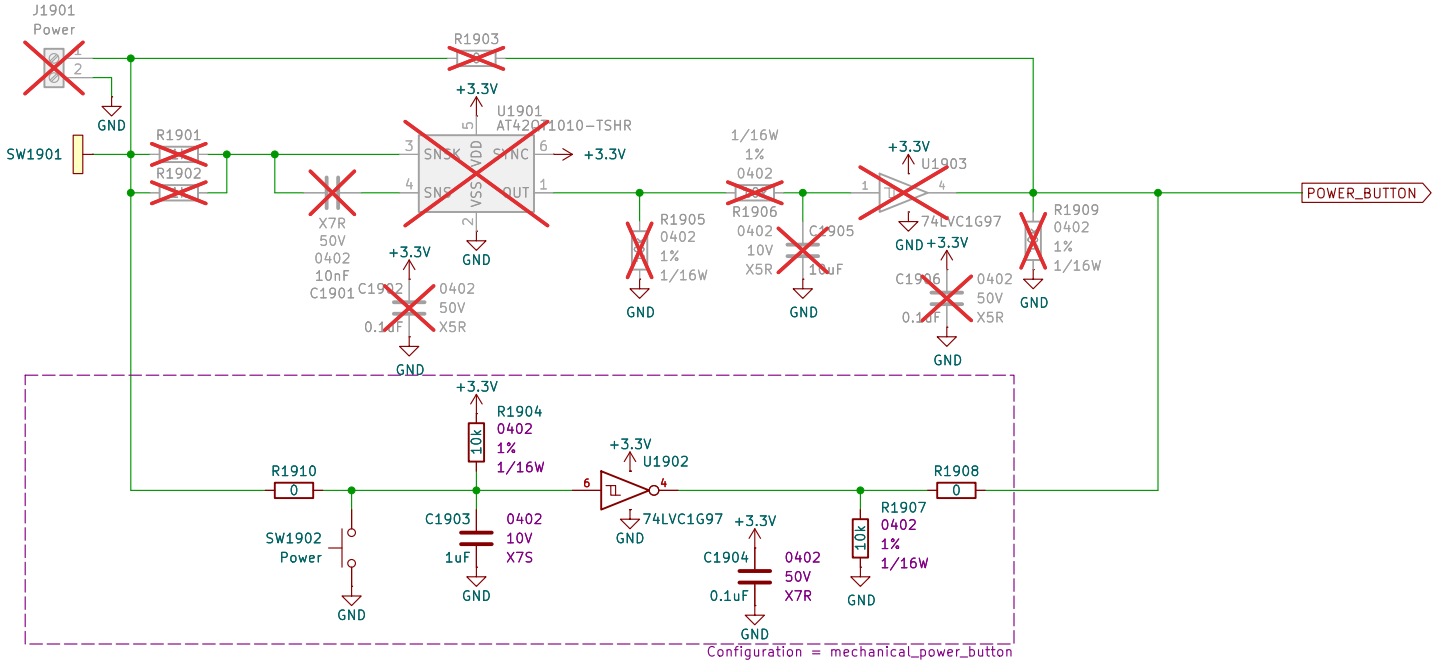
Drew Maatman and Drew Koetting

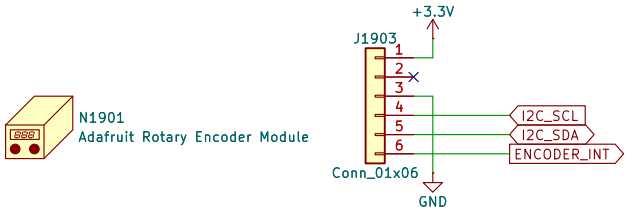
Sheet: /LED Panel Buffers/

File: led_panel_buffers.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 18/21





I2C 7-Bit Address:
0x36

Drew Maatman and Drew Koetting

Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 19/21

H2001
MountingHole

H2003
MountingHole

H2002
MountingHole

H2004
MountingHole

Screw
MK2001

Screw
MK2005

Screw
MK2002

Screw
MK2006

Standoff
MK2003

Standoff
MK2007

Standoff
MK2004

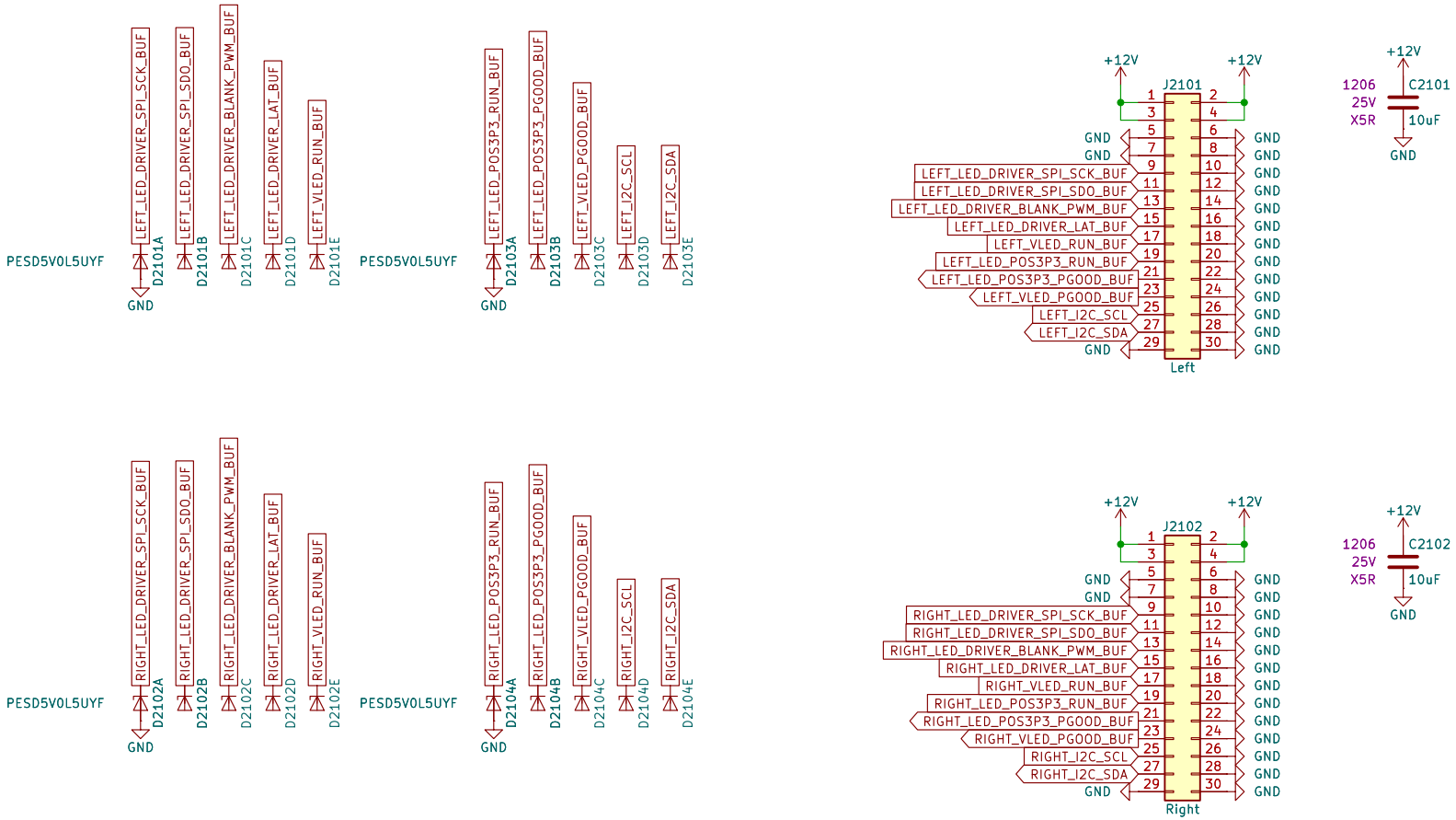
Standoff
MK2008

Drew Maatman and Drew Koetting

Sheet: /Mechanical/
File: mechanical.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 20/21



Drew Maatman and Drew Koetting

Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 21/21